

Current

1289 - floatToHalfKernel

Size

(884736, 1, 1)x(256, 1, 1)

Time

10.78 ms

Cycles

20,704,778

GPU

0 - NVIDIA GeForce RTX 4060 Laptop GPU

SM Frequency

1.92 Ghz

Process

[35348] MPI.exe

Attributes

Summary

Details

Source

Context

Comments

Raw

Session

Compare

Tools

View

Export

This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [%]	Function Name	Demangled Name	Duration (3,284.2)	Runtime Improvement [ms] (1,013.50 ms)	Compute Throughput	Memory Throughput [	# Registers [register/t	Grid Size
0	38.04	floatToHalfKernel	floatToHalfKernel...	10.78	4.10	61.96	49.62	24	884736,
1	50.00	floatToHalfKernel	floatToHalfKernel...	0.01	0.00	21.04	21.59	24	160,
2	47.06	preprocessNormali...	preprocessNormali...	0.04	0.02	50.13	14.58	16	3888,
3	23.78	matrixMulTiled	matrixMulTiled..flo...	535....	127.38	76.22	7.80	32	256,
4	43.78	addBias	addBias(float *, con...	0.02	0.01	56.22	25.98	16	1152,
5	41.28	reluKernel	reluKernel(float..	0.01	0.01	41.95	34.75	24	1152,
6	91.67	matrixMulTiled	matrixMulTiled..flo...	7.21	6.61	6.59	0.68	32	1,
7	95.83	addBias	addBias(float *, con...	0.00	0.00	0.21	0.62	16	1,
8	47.17	preprocessNormali...	preprocessNormali...	0.04	0.02	50.02	14.55	16	3888,
9	23.79	matrixMulTiled	matrixMulTiled..flo...	535....	127.49	76.21	7.80	32	256,
10	44.16	addBias	addBias(float *, con...	0.02	0.01	55.84	25.79	16	1152,
11	41.42	reluKernel	reluKernel(float..	0.01	0.01	42.13	34.92	24	1152,
12	91.67	matrixMulTiled	matrixMulTiled..flo...	7.22	6.61	6.59	1.11	32	1,
13	95.83	addBias	addBias(float *, con...	0.00	0.00	0.21	0.64	16	1,
14	48.51	preprocessNormali...	preprocessNormali...	0.02	0.01	49.73	14.84	24	972,
15	32.27	matrixMulTiledFP16	matrixMulTiledFP1...	529....	170.99	67.73	15.01	48	256,

The following performance optimization opportunities were discovered for this result. Follow the rule links to see more context on the Details page.
Note: Speedup estimates provide upper bounds for the optimization potential of a kernel assuming its overall algorithmic structure is kept unchanged.

Wait Stalls

Est. Speedup: 38.04%

On average, each warp of this workload spends 13.7 cycles being stalled waiting on a fixed latency execution dependency. Typically, this stall reason should be very low and only shows up as a top contributor in already highly optimized kernels. Try to hide the corresponding instruction latencies by increasing the number of active warps, restructuring the code or unrolling loops. Furthermore, consider switching to lower-latency instructions, e.g. by making use of fast math compiler options. This stall type represents about 52.4% of the total average of 26.2 cycles between issuing two instructions.

Local Memory Usage

Est. Speedup: 20.03%

This workload accesses local memory, accounting for 57.14% of all sectors requested in L1TEX. Local memory refers to a memory space that is private to each thread. Although it is logically distinct, local memory resides in global memory. This can lead to higher memory latency compared to the usage of registers or shared memory, a decrease in available bandwidth to

Key Performance Indicators